SRII AKHILLESSH GADDAM

DATA ENGINEER | ETL PIPELINE DEVELOPER | GCP & AWS

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Work authorization: OPT through 2029

PROFESSIONAL SUMMARY

Data Engineer with 4+ years designing, building, and scaling production **batch and streaming ETL/ELT data pipelines** on **GCP and AWS** using **Python, SQL, Apache Spark, Databricks, Kafka, BigQuery, Snowflake, and Airflow**. Experienced in data architecture and **lakehouse design with Delta Lake, Databricks Workflows, and Unity Catalog**, plus **data modeling, CDC, and incremental processing**. Cut a daily ETL runtime by **75% (2 hours to 30 minutes)** through Spark performance optimization. Hands-on with data quality, automated data validation, data governance and security, **Terraform** infrastructure as code, **Git-based CI/CD**, and cloud-native data platforms that power analytics, BI, machine learning, and AI applications in client-facing consulting environments. M.S. in Computer Science (2026).

TECHNICAL SKILLS

Programming & Development: Python, PySpark, SQL, Spark SQL, Apache Spark, pandas

Data Engineering: ETL/ELT, Batch & Streaming Pipelines, CDC, Incremental Loads, Data Modeling, Dimensional Modeling, Data Architecture, Data Integration, Data Warehousing, Production-Grade SQL, Lakehouse Architecture, Data Validation & Cleansing, Source-to-Target Reconciliation, Query Optimization, Partitioning & Clustering, Spark Optimization, Performance Tuning

Big Data & Streaming: Apache Spark, Databricks, Delta Lake, Databricks Workflows, Unity Catalog, Apache Kafka, Spark Streaming, Google Cloud Dataflow, Distributed Data Processing, Event-Driven Processing

Orchestration & Transformation: Apache Airflow, Cloud Composer, Dataform, Databricks Workflows

Cloud & Data Platforms: GCP (BigQuery, Cloud Composer, Dataflow, Cloud Run), AWS (S3, Redshift, Lambda, CloudWatch), Snowflake, SQL Server

Data Governance & Quality: Data Lineage, Metadata Management, Access Control & Security, RBAC, Data Quality Monitoring, Alerting, Root Cause Analysis

DevOps & Delivery: Terraform (Infrastructure as Code), GitHub Actions, CI/CD, Git, Version Control, Pull Requests, Code Reviews, Testing, SDLC, Agile/Scrum

AI & Analytics: RAG Pipelines, Semantic Search & Retrieval, Vector Databases (ChromaDB), Embeddings, FastAPI, REST APIs & Integrations, Dashboards, ML Data Pipelines

PROFESSIONAL EXPERIENCE

Data Engineer (Contract) | GCP Data Engineering, Client Analytics

OneMagnify

May 2025 – Present

- Designed, built, and maintained **15–20 SQL-first batch pipelines on GCP** using **BigQuery, Dataform, Cloud Composer/Airflow, and Dataflow**, delivering ~40 GB of curated data daily to 3 client analytics teams.
- **Automated end-to-end client data delivery**, from ingestion and validation through **BigQuery and dashboards**, reducing manual effort and accelerating report availability.
- Modeled **dimensional, analytics-ready datasets** in BigQuery and Dataform with reusable, version-controlled transformation layers, giving BI, analytics, and ML teams one consistent source of truth.
- Optimized **complex BigQuery SQL** using **partition pruning and clustering**, cutting query runtime by **25%** and bytes scanned by **30%** and keeping processing costs predictable as client data grew.
- Orchestrated dependencies, scheduling, retries, and failure alerts in **Cloud Composer (Airflow) DAGs**, keeping daily client data deliveries on schedule.
- Built **3 API integrations** as Python services on **Cloud Run** and **Dataflow** ingestion jobs that standardize third-party and platform data, consolidating it into a single trusted analytics environment in BigQuery.
- Owned **automated data validation and cleansing pipelines** and traced failures upstream to their root causes, raising scheduled pipeline success rates from **96% to 99%** over three months.
- Partnered with data scientists to build **GCP data integrations that feed AI/ML models**, helping move models from experimentation into **production workflows**.
- Translated client and stakeholder requirements into **8 documented, curated datasets and 3 dashboards**, shortening the path from business question to insight.
- Shipped changes through **Git, pull requests, code reviews, and GitHub Actions CI/CD**, and turned recurring ad-hoc requests into **4 reusable data assets**, saving **~5 team hours weekly**.

Data Engineer | AWS & Databricks Data Engineering

LTIMindtree

August 2022 – July 2024

- Built and maintained **12 batch and 3 streaming ETL/ELT pipelines on AWS** with **Python, SQL, PySpark, Spark SQL, Databricks, Apache Kafka, S3, Redshift, and Lambda**, processing ~80 GB of structured data daily for client reporting and analytics.
- Reduced daily ETL runtime by **75% (2 hours to 30 minutes)** by implementing **incremental processing** and optimizing Spark joins and partitioning, making reports available earlier each day.
- Designed a scalable **Databricks lakehouse** on **Delta Lake**, orchestrated with **Databricks Workflows** and governed with **Unity Catalog**, centralizing data access, security, metadata, and lineage.
- Implemented **change data capture (CDC)** from Kafka event streams into Delta Lake tables, so downstream reports reflected source changes in near real time.
- Loaded curated datasets into **Amazon Redshift and Snowflake** data warehouses, giving enterprise reporting teams query-ready data.
- Provisioned cloud data infrastructure with **Terraform (infrastructure as code)**, cutting environment setup from **2 business days to 4 hours** with consistent, reproducible environments.
- Implemented **CloudWatch monitoring, alerting, and SQL/Python data-quality checks**, reducing average incident resolution time from **90 to 60 minutes**.
- Built **GitHub Actions CI/CD workflows** across the SDLC for 5 data applications, cutting deployment time from **45 to 15 minutes**, and set shared engineering standards with product and domain teams.

Data Miner Intern | ETL & Data Migration

ADP

January 2022 – July 2022 | Hyderabad, India

- Built **SQL Server and Python (pandas) ETL workflows** to migrate **~80K employee records across 4 clients** into ADP Lyric HCM, enabling payroll processing on the new platform.
- Transformed data in **SQL Server staging tables** into Lyric import files, using pandas to split and merge source datasets to match the target load format.
- Mapped employee, position, and work-agreement records to Lyric's **position-management data model**, including multi-agreement employees, so every record loaded with the correct structure.
- Developed **data-quality and validation checks** (special-character cleanup, source-to-target reconciliation), ensuring accurate payroll loads.

PROJECTS

LTU Sky Chatbot – Hybrid RAG Data Pipeline | *Lawrence Technological University*

- Built a **data ingestion pipeline** that scrapes unstructured LTU faculty and admissions web data with **Playwright** and indexes it in a **ChromaDB vector database** and a **BM25** keyword index, turning scattered web pages into searchable knowledge.
- Implemented **hybrid retrieval** combining BM25 and semantic search with **Reciprocal Rank Fusion**, returning more relevant answers than keyword or vector search alone.
- Served the chatbot through a **FastAPI** backend and Next.js frontend with **local LLMs (Ollama phi3/mistral)**, keeping university data on-premises, and built **530 Q&A pairs** to evaluate answer quality.

rACTor Autonomous Vehicle – Self-Driving Perception & Navigation (ROS2) | *1st Place, IGVC 2026 (Self-Drive)*

- Built a **ROS2 Python package** fusing **time-synchronized LiDAR and camera streams** (YOLO detection + VLP-16 point-cloud filtering) across a split-compute Raspberry Pi 5 and laptop setup.
- Built a **real-time data pipeline from the rACTor vehicle to the cloud**, streaming camera and LiDAR sensor data to build training datasets and train **YOLO** models for **path recognition and obstacle detection**.
- Trained and tuned **obstacle avoidance and path tracking** using **Nav2 and RViz**, running **thousands of simulations** so the vehicle stays on the correct path, avoids obstacles, and follows traffic rules; team won **1st place at IGVC 2026 (Self-Drive) with a perfect score**.

EDUCATION & CERTIFICATIONS

Master of Science, Computer Science | GPA: 3.6

Lawrence Technological University, Southfield, MI | 2024 – 2026

Bachelor of Technology, Computer Science and Engineering | GPA: 3.5

Sreenidhi Institute of Science and Technology, Ghatkesar, Telangana, India | 2018 – 2022

Certifications: Databricks Certified Data Engineer Associate • AWS Certified Data Engineer – Associate